

Detecting cognitive engagement in online course forums: A review of frameworks and methodologies

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ABSTRACT

A key aspect of online learning in higher education involves the utilization of course discussion forums. Assessing the quality of posts, such as cognitive engagement, within online course discussion forums, and determining students' interest and participation is challenging yet beneficial. This research investigates existing literature on identifying the cognitive engagement of online learners through the analysis of course discussion forums. Essentially, this review examines three educational frameworks - *Van Der Meijden's Knowledge Construction in Synchronous and Asynchronous Discussion Posts (KCSA)*, *Community of Inquiry (CoI)*, and *Interactive, Constructive, Active, and Passive (ICAP)*, which have been widely used for students' cognitive engagement detection analyzing their posts in course discussion forums. This study also examines the natural language processing and deep learning approaches employed and integrated with the above three educational frameworks in the existing literature concerning the detection of cognitive engagement in the context of online learning. The article provides recommendations for enhancing instructional design and fostering student engagement by leveraging cognitive engagement detection. This research underscores the significance of automating the identification of cognitive engagement in online learning and puts forth suggestions for future research directions.

1. Introduction

Online education has drawn considerable attention from researchers due to its flexible and accessible delivery method for online learners. A key component of effective online learning is cognitive engagement, which refers to the psychological investment and effort made by the learner to understand what is studied and to achieve a higher level of comprehension in a specific area of study (Kew and Tasir, 2021). Improving cognitive engagement can help students develop critical thinking, move from procedural tasks to problem-solving tasks, and think about the learning process (Lv et al., 2022). With cognitive engagement detection, it is possible to gain insightful knowledge about students' learning processes and inform instructional support for better results (Kew and Tasir, 2021).

This study presents a narrative review of educational frameworks and artificial intelligence techniques used to detect students' cognitive engagement analyzing course discussion forums. Students in online learning engage in collaborative learning, share ideas and comments, and develop their critical thinking through course discussion forums. However, determining cognitive engagement through students' posts is challenging due to the absence of any immediate visual and

auditory cues, variations in how students' express engagement, and not identifying their internal cognitive activities. Additionally, assessing diverse learning objectives and the role of social and affective aspects made it difficult to gauge students' cognitive engagement accurately. Some educational frameworks, such as *Community of Inquiry (CoI)* (Akyol et al., 2009), *Interactive, Constructive, Active, and Passive (ICAP)* (Chi, 2009), and *Van Der Meijden's knowledge construction in synchronous and asynchronous discussion posts (KCSA)* (Meijden, 2005), have been widely used by the researchers for detecting cognitive engagement of online learners in their course discussion forums. Along with the above frameworks, the recent advancement of machine learning, deep learning, text mining, natural language processing, statistical analysis, and large language models have made this possible to automate the analysis of cognitive engagement in course discussion forums.

This paper presents a narrative review of the existing works on identifying online learners' cognitive engagement through the examination of course discussion forums using the *CoI*, *ICAP*, and *KCSA* frameworks, machine learning and natural language processing (NLP) techniques. We aim to advance the knowledge of cognitive engagement detection

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methods in online learning by synthesizing the existing literature and examining the following four research questions: (1) *What theoretical frameworks are widely used for students' cognitive engagement detection by analyzing course discussion posts?* (2) *How are these frameworks employed to code discussion posts to identify cognitive engagement in online courses?* (3) *What machine learning (ML) and artificial intelligence (AI) techniques have been used with the cognitive engagement frameworks for cognitive engagement detection?* (4) *What are the pros and cons of using ML and AI techniques with the cognitive engagement frameworks for cognitive engagement detection using forum posts analysis?* We expect that this review will further advance the knowledge of cognitive engagement analysis in online courses and offer useful information for educational institutions and academics in creating productive online courses and smart learning environments. We also hope to offer guidance to researchers, teachers, and practitioners on how to use the cognitive engagement detection frameworks for effective instructional design and learner support.

The rest of the paper is organized as follows. *Section 2* presents an in-depth discussion on CoI, ICAP and KCSA frameworks and their coding schemes from the perspective of students' cognitive engagement detection in online learning. *Section 3* explores the recent research studies on cognitive engagement detection and explores how machine learning models and NLP techniques have been used along with the various educational frameworks for cognitive engagement detection and use it to improve students' learning outcome. *Section 4* discusses several open-ended research issues related to cognitive engagement detection that we revealed from existing literature. Finally, *Section 5* concludes the review with some insightful discussions and future recommendations.

2. Educational frameworks for cognitive engagement detection

In an online learning environment, cognitive engagement refers to the level of active participation of the online learners in the learning process and intellectual engagement with their online courses. Cognitive engagement can be promoted in online learning in various ways, such as interactive course materials, discussion boards, and cooperative group activities. Cognitively engaged students are more likely to comprehend the information better, retain it for longer, and eventually perform better academically (Kew and Tasir, 2021). The importance of measuring students' cognitive engagement during online learning cannot be overstated. Cognitive engagement detection aids instructors in various ways in evaluating the efficiency of their lesson plans and online learning platforms. By keeping track of cognitive engagement, instructors can assess both the level of student participation in the learning process and the efficacy of their instructional approaches. Instructors might spot learners who could be having trouble or are not interested in the subject matter. With the aid of this information, educators can offer these learners extra assistance or resources to help them be active. Learners deeper learning and critical thinking can be encouraged with the aid of cognitive engagement detection. Instructors can assist students in honing their problem-solving and analytical abilities, which are crucial for success in many disciplines, by monitoring and encouraging cognitive engagement. The detection of cognitive engagement can yield insightful information for course assessment and development. Instructors can pinpoint course components that may need improvement and make changes to improve student learning results by analyzing cognitive engagement data. However, measuring cognitive engagement in online learning is challenging, as it involves both observable behaviors, such as participation in online discussions, and unobservable cognitive processes such as critical thinking and problem-solving. However, various educational frameworks provide structured ways to evaluate cognitive engagement, by focusing on aspects of cognitive presence and interactive learning activities. As we mentioned earlier, some such frameworks are ICAP, CoI and KCSA.

2.1. ICAP framework

The ICAP framework (Chi, 2009) focuses on how the students cognitively engage with their learning activities. This framework basically focuses on four types of engagement: *interactive*, *constructive*, *active*, and *passive*. *Interactive* engagement includes activities where students actively collaborate and communicate with one another as well as the instructor. Discussing with peers or responding to questions online discussion boards, collaborative projects, and peer review exercises are some examples of *interactive* engagement. A higher level of cognitive engagement is typically found in this category, where students actively interact with the instructor, their peers, and each other to deepen their knowledge about the course topic. *Constructive* engagement focuses on the tasks that require students to build something fresh out of the course material (e.g., organizing information or summarizing key points, producing a presentation, an essay, or a project plan). As students actively apply and synthesize the information to create something new, constructive activities are thought to have high levels of cognitive engagement. The *Constructive* category is further divided into two subcategories: *C1* and *C2*. Students in *C1* subcategory display reasoning skills by elaborating on points, explaining phenomena, or providing in-depth analysis related to the course content. Students in *C2* subcategory go beyond the course materials and propose new ideas or content related to the course, demonstrating creativity and original thinking.

The *Active* category focuses on the exercises where students actively engage with and process the course material (e.g., making notes or coming up with examples, taking notes, responding to inquiries, or resolving issues). Given that students are actively processing and manipulating information, active activities are thought to have moderate levels of cognitive engagement. The *Active* category is divided into two subcategories: *A1* and *A2*. Students in *A1* subcategory engage with course materials actively, showing involvement and interaction with the content. In contrast, students in *A2* subcategory are active, but they do not specifically mention course content. Their engagement might be more general or unrelated to the course material. *Passive* category focuses on activities where students are passively taking in knowledge by observing or listening (e.g., reading or watching video lectures without actively processing the information). Because learners are not actively processing or interacting with the information, passive activities are thought to have low levels of cognitive engagement. Finally, an additional category is included in this framework which is called "*Other*" which encompasses behaviors that do not fit into the above-defined categories.

For detecting cognitive engagement using ICAP framework analyzing course discussion forum, researchers use a coding scheme which is shown in Fig. 1. This coding scheme provides a structured way to assess and categorize student engagement behaviors, allowing educators and researchers to gain insights into online courses. ICAP framework can be used to identify these areas in online courses and encourage deeper learning and engagement among students. The ICAP framework allows instructors to obtain knowledge about how their students interact with the course materials and activities, and to use this knowledge to modify their course design and teaching process.

2.2. CoI framework

The CoI framework (Akyol et al., 2009) emphasizes the social and cognitive aspects of the learners. It emphasizes social presence, cognitive presence, and teaching presence as the three key components of effective online learning. The CoI framework is used for understanding and assessing the quality of students' learning. The framework has been used to evaluate online learning environments. As we mentioned earlier, this framework comprises three interrelated elements that are essential for achieving a successful online learning experience: *Social*

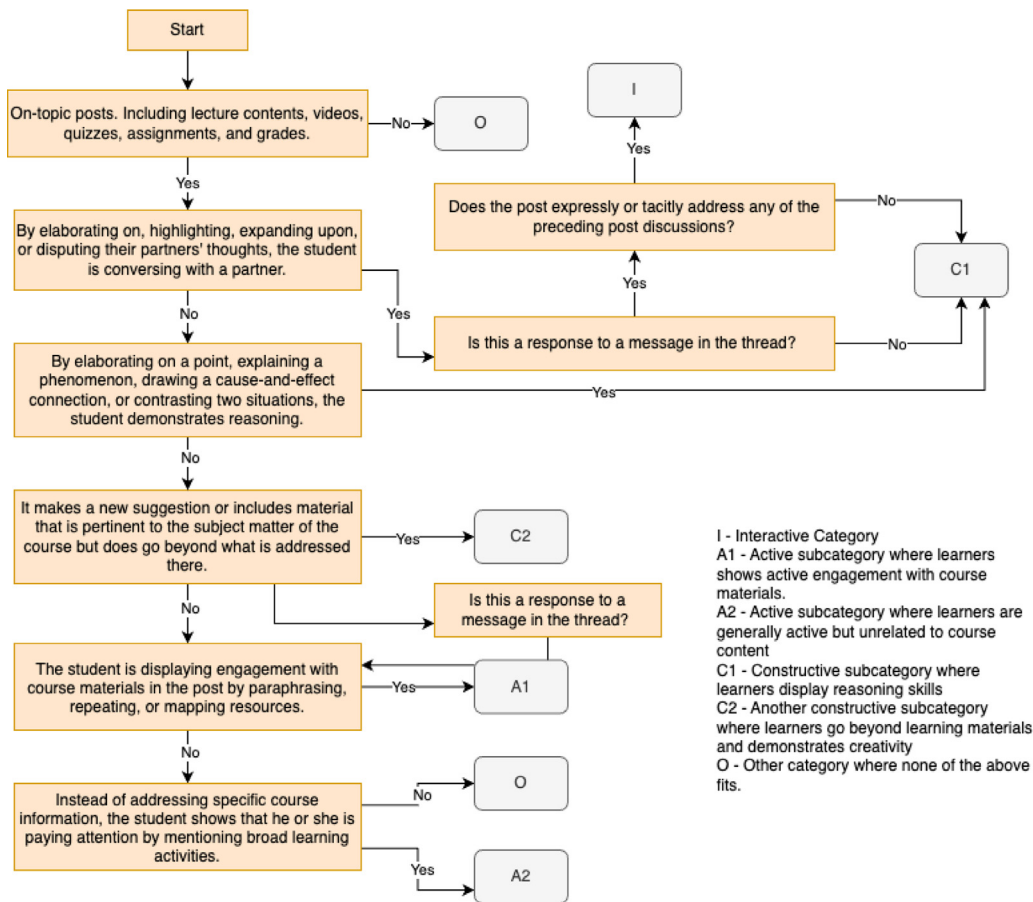


Fig. 1. Coding scheme of ICAP framework.

Presence, Cognitive Presence, and Teaching Presence. *Social Presence* describes how connected and involved students feel with their instructors, peers, and other students in an online learning environment. This covers elements like the usage of communication tools, the degree of participant participation, and how much the learning experience is customized. *Cognitive Presence* represents the level of a learner's ability to make sense of the course material and to engage in critical analysis and research. This involves problem-solving exercises, data analysis, and evidence evaluation, as well as the utilization of reflection exercises and group work among students. *Teaching Presence* represents the level to which the instructor creates and supports the learning environment to support social and cognitive presence. This involves tasks like giving learners clear instructions and feedback, organizing conversations, and offering support and direction.

The CoI framework is used to measure the efficacy of online learning environments by rating the quality of the above three components. Educators and researchers can use the framework to identify the advantages and disadvantages of online course design as well as to direct the creation of strategies to improve the learning process. The CoI framework has been widely used in research studies to assess how well online learning environments support learners' cognitive involvement. One study by Gray and DiLoreto (2016) has demonstrated, for instance, that courses with higher levels of social and cognitive presence are more successful at encouraging learners to engage in deep learning and critical thinking. The CoI framework is a useful resource for comprehending and evaluating the quality of online learning environments, as well as for developing and accessing efficient methods for encouraging cognitive engagement among students.

The coding scheme for CoI framework is shown in Table 1. In CoI framework, the first component, *cognitive presence*, represents the

degree to which students create and validate meaning through prolonged contemplation and conversation that places an emphasis on helping them develop critical and higher-order thinking skills (Garrison et al., 1999). *Cognitive presence* comprises the four dimensions in its learning cycle: triggering event (conceptualization), (b) exploration (thought production), (c) integration (knowledge synthesis), and (d) resolution (application of knowledge and vicarious testing). The second component, *social presence*, represents the growth of social interactions between learning assemble people while preserving an advantageous social environment. *Social presence* has three dimensions which include *affective expression*, *course organization*, *acknowledge receipt*, and *group cohesion*. The third component, *teaching presence*, describes course organization and design, direct instruction, facilitation, and the instructor's role both before and during teaching. *Teaching presence* plays an important role in enhancing student pleasure, perceived learning, and community sense.

2.3. KCSA framework

The KCSA framework is used for assessing the quality of online discussions in terms of the cognitive processes that the students engage in course discussion forums. This framework uses four stages of students' cognitive engagement in online discussions, which include *exploration*, *conceptualization*, *application*, and *reflection*. At the *exploration* stage, students explore the topic of discussion by brainstorming ideas, asking questions, and sharing their initial thoughts and experiences. This stage is characterized by open-ended and exploration discussions, and the emphasis is on generating a wide range of ideas. At the *conceptualization* stage, students begin to develop a shared understanding of the topic of discussion by clarifying concepts, defining terms, and identifying patterns and themes. This stage is characterized by more focused and

Table 1
Coding scheme of CoI framework.

Elements	Categories	Indicators
Cognitive presence	Triggering event	New subject, message expresses student's confusion.
	Exploration	Although messages lack a logical conclusion, they contain information regarding the source of the misunderstanding.
	Integration	Connecting ideas, Messages make logical, well-supported recommendations about how to address the situation.
	Resolution	Applying new concepts, messages frequently take the form of new constructions and debate, test, or apply the earlier conclusions.
Social presence	Affective expression	Apology, Thank you, Emotions.
	Group cohesion	Social, Greeting, Encouraging, and Community Building
	Course organization	Technical, class management, time out
	Acknowledge receipt	Nodding
Teaching presence	Design and organization	Time out, technical, course management, class management
	Facilitating discourse	Clarification, specific query, giving task, confirmation, understanding, direction
	Direct instruction	Explanation of content, additional explanation definition.

structured discussions, and the emphasis is on developing a deeper understanding of the topic. At the *application* stage, students apply what they have learned about the subject to real-world problems. The emphasis at this level is on making connections between abstract concepts and actual circumstances, which is demonstrated using examples and case studies. At the *reflection* stage, students analyze their comprehension and reasoning as they consider the lessons that have been learned during the conversation. The focus of this stage is on gaining a deeper understanding of one's thought processes and is characterized by metacognitive processes including self-evaluation and self-reflection.

The coding scheme for the KCSA framework is shown in [Table 2](#). This coding scheme is used to examine online discussions and determine the level of cognitive presence of the students in an online learning environment. Researchers often use this coding scheme to rate the effectiveness of online discussions and identify areas where online learning environments can be improved. KCSA categorizes the students' learning activity as *cognitive* (asking questions, providing answers, and providing information), *affective*, *regulative*, and *rest*. However, the *affective*, *regulative*, and the *rest* are not considered when determining the student's level of cognitive involvement because these categories do not entail any cognitive components.

Within the *cognitive* category, the data for question-asking includes questions with/without explanation and requests for agreement (such as CHV1, CHV2, and CHVER), whereas the data for answer-giving includes replies with/without explanation (such as CHG1 and CHG2). Giving information includes providing information with or without elaboration, making references to earlier information, assessing the content, accepting the statement with or without elaboration, and rejecting the statement with or without elaboration (for example, CI 1, CI2, CIT, CIE, ACCEPT-, ACCEPT+, REJECT- and REJECT+). These schemes are used to identify the level of cognitive engagement of online students.

2.4. Some key differences among ICAP, CoI, and KCSA frameworks

Some key differences among ICAP ([Chi, 2009](#)), CoI ([Akyol et al., 2009](#)), and KCSA ([Meijden, 2005](#)) frameworks are summarized in [Table 3](#). These three frameworks provide a structured way for analyzing student behavior and modifying teaching techniques to improve student learning outcomes. These frameworks can assist instructors in identifying the cognitive involvement of students during online learning. Identifying cognitive engagement is essential for instructors to encourage student learning, to spot students who are having difficulty in learning, and to enhance the standard of online learning environments. Educational researchers widely use these frameworks to enhance pedagogical approaches and evaluate the efficacy of teaching strategies in many learning contexts.

3. Methods for cognitive engagement detection analyzing course discussion posts

Researchers have devoted significant effort to understanding and detecting cognitive engagement in online learning environments, using diverse frameworks and techniques to analyze and interpret students' cognitive activities. This section provides a comprehensive review of the literature on cognitive engagement detection, with a focus on analyzing course forums using artificial intelligence, machine learning, and NLP techniques. It also examines the previously discussed three widely recognized educational frameworks, ICAP, CoI, and KCSA, which offer valuable perspectives for exploring the depth and quality of learners' engagement. These frameworks illuminate the dynamic interplay between cognitive processes and online learning experiences.

3.1. Methods for cognitive engagement detection using ICAP framework

In this subsection, we discussed noteworthy studies that utilizes ICAP framework for cognitive engagement detection. These studies employed machine learning and deep learning methods in conjunction with the ICAP framework and NLP to automatically categorize and analyze student engagement, providing valuable insights into online learning environments and pedagogical strategies.

[Hayati et al. \(2020\)](#) presented an approach that analyses students' social interactions in online discussion forums using text-mining and predicted learners' cognitive involvement. This approach incorporated the ICAP framework to investigate and classify whether a student is a passive, active, constructive, or interactive participant based on a corpus of posts in a software engineering course delivered through an online learning platform. The authors used SVM for classifying the learners' cognitive engagement into the above four categories of the ICAP framework. The obtained findings showed intriguing precision with a higher classification accuracy and Cohen Kappa, demonstrating the effectiveness of the suggested technique with a high accuracy.

[Liu et al. \(2022\)](#) presented a text classification model to automatically detect students' emotional and cognitive engagement and investigate their relationships with the students' achievement. This method used an interpretable NLP model called BERT-CNN (bidirectional encoder representation from the transformers-convolutional neural network) ([Devlin et al., 2018](#)) to analyze learners' discussions in a MOOC forum ([Galikyan et al., 2021](#)). In this study, authors found that the co-occurring emotion and cognition indicators have a combined effect on learning achievement. It also offered insights for enhancing MOOC completion rates and learning outcomes, such as introducing engaging and meaningful discussion tasks to evoke positive or thought-provoking emotions.

Table 2
Coding scheme for KCSA framework.

Asking questions	CHV1	The student inquiries about a fact or just asks what the answer is. The question does not provoke any explanation
	CHV2	Posing inquiries that demand an explanation (comprehension or explication)
	CHVER	The student poses a question to determine whether other people concur with his or her statement, conclusion, or response.
Giving answers	CHG1	The student responds to the question but does not elaborate or give more details.
	CHG2	The student answers the question elaborately and gives more explanation.
Giving information	CI1	Giving information (a concept or idea) without more explanation
	CI2	The student formulates an idea or thought and then explains or elaborates on it.
	CIT	The student refers to earlier remarks or previously formulated information.
	CIE	The student assesses, paraphrases, or concludes the materials.
	ACCEPT-	The student acknowledges another participant's contribution without adding anything additional.
	ACCEPT+	The student declines to acknowledge the contribution of a participant and gives no more justification.
	NAACCEPT-	The student declines to acknowledge the contribution of a participant and gives no more justification.
	NACCEPT+	The student refuses to acknowledge the contribution of a participant and gives elaboration for refusal.

Table 3
Some key differences among ICAP [Chi \(2009\)](#), CoI [Akyol et al. \(2009\)](#), and KCSA [Meijden \(2005\)](#) frameworks.

Category	Differences
Focus	ICAP focuses on the learning processes, whereas CoI focuses on the social processes of learning. KCSA, on the other hand, focuses on the cognitive processes involved in learning.
Coding scheme	ICAP uses a coding scheme that classifies student interactions into four categories: interactive, cognitive, active, and passive. CoI categorizes student interactions into three types: cognitive, social, and teaching presence. KCSA categorizes student interactions into five categories: cognitive, metacognitive, affective, social, and strategic.
Application	CoI has been used in a variety of learning contexts while ICAP has been primarily used in STEM education. Both conventional classroom settings and online learning environments make use of the KCSA framework.
Level of granularity	While CoI framework offers a more in-depth look at social processes, the ICAP framework offers a broad overview of student engagement. A more thorough examination of cognitive processes is offered by the KCSA framework.

[Atapattu et al. \(2019\)](#) employed the ICAP framework to measure cognitive engagement in a professional development MOOC. Using word embeddings, they automated the identification of instructors' community contributions as either "active", involving the manipulation of course materials, or "constructive", involving the generation of new knowledge. They also explored individual variations in engagement across different units. Additionally, the study incorporated a manual content analysis to examine constructive contributions. Out of 67 cases analyzed, all but one were identified as containing "constructive knowledge", providing a robust foundation for replicating their methodology to analyze cognitive engagement in community-centric MOOC models. Their findings highlighted that participants' cognitive engagement is influenced by the nature of MOOC tasks.

[Liu et al. \(2023b\)](#) presented a semi-supervised cognitive engagement classification method called B-LIWC-UDA. This method incorporated BERT ([Devlin et al., 2018](#)) and LIWC ([Pennebaker et al., 2001](#)) cognitive lexicon dual feature embeddings. The method uses advanced data-augmentation methods and fuses generic semantic features and implicit cognitive features to optimize its performance ([Moore et al., 2019](#); [Wen et al., 2014](#)). The results of this study showed that the cognitive features taken from the LIWC cognitive lexicon could efficiently characterize cognitive engagement for both the ICAP and the CoI frameworks. The other method, B-LIWC-UDA, used an advanced data augmentation module called EDA ([Wei and Zou, 2019](#)) that serves as a noise insertion approach that can produce diverse and effective augmented samples without changing the labels of the original data samples. The B-LIWC-UDA outperformed the other models in major evaluation metrics, such as accuracy and resource consumption. This method is found to be useful to help instructors monitor learners' learning status and intervene earlier with at-risk learners, potentially increasing online learning retention by investigating cognitive engagement in forum discussions.

[Gorgun et al. \(2022\)](#) presented a predictive model for automatically identifying the cognitive engagement in online discussion posts by employing ICAP and Bloom's taxonomy. Authors used the ICAP coding scheme to label students' posts for capturing their level of cognitive engagement. The labeled data were used to train three machine learning classifiers — decision tree (DT), random forest (RF), and support vector machine (SVM), where the SVM outperformed the other classifiers. The features for the classifiers were extracted by applying CohMetrix to student posts and non-linguistic contextual features (e.g., number of replies). Authors planned to explore some new features, such as AWL ([Coxhead, 2000](#)) and Flesh-Kincaid ([Kincaid et al., 1975](#)) in their future study. The above studies showed that automated models in conjunction with the ICAP framework hold promises for effectively categorizing and predicting cognitive engagement in online learning.

3.2. Methods for cognitive engagement detection using CoI framework

In this subsection, we examined methods that used NLP and machine learning models in conjunction with the CoI framework for detecting cognitive engagement. Numerous studies have explored various aspects of cognitive engagement in online learning environments, emphasizing its critical role in shaping effective pedagogy and improving student learning outcomes within the CoI framework.

[Lee et al. \(2022\)](#) proposed a method to automatically measure and enhance the quality of interactions in discussion forums. This study presented a machine learning model that can be used to identify the phases of cognitive presence exhibited by a student in their forum posts. It also suggested several future applications of such a model to help online students develop higher-order thinking. The authors collected discussion posts from two online courses which used Piazza ([Piazza, 2024](#)), an online discussion tool. The posts were manually coded based

on the CoI coding scheme in order to explore trends in cognitive presence within and across the courses. The authors further used this coded data to analyze the relationship between students observed cognitive presence and course grades. In terms of testing and building an ML model, this study used BERT model to train with a large text corpus and fine-tune the model. The results suggest that deeper cognitive engagement with course concepts, as expressed by higher cognitive presence, is associated with better learning outcomes for students in both course settings. This research study is expected to have a beneficial impact on online instructors or curriculum developers in higher education.

Villanueva et al. (2022) looked at how social and teaching presences interacted in blended learning in K–12 settings. This study examined three blended learning classes based on data collected through surveys, student focus group discussions, teacher interviews, class observations and archived data. When participating in reflective community-building collaborative activities, students gave excellent marks to their perceived learning at the integration and resolution levels of the cognitive presence. Due to their shared metacognition, which is defined as “an awareness of one’s learning in the process of constructing meaning and creating understanding associated with self and others” (G. DR, 2017), students’ self-regulation and co-regulation techniques were impacted by group activity. The students are held more accountable for taking on their obligations and learning time management. This study demonstrated the development of learning communities and the relevance of the CoI in the K–12 educational setting.

Ozogul et al. (2022) investigated how to promote cognitive presence in an online graduate course. To gauge cognitive presence, the authors employed CoI for self-reporting and Linguistic Inquiry and Word Count (LIWC) software for actual behavior tracking. This study also looked at how cognitive presence interacted with the behavioral presence in an educational setting. Students perceived and demonstrated a higher level of cognitive presence in their discussion board behaviors, according to the findings. Results also indicated that the perceived cognitive presence is strongly predicted by teaching and social presence. Authors also argued that the instructor’s responsiveness in discussion posts and the creation of dialogue, the creation of course assignments as online hands-on projects, the interviewing of guest speakers on course topics, weekly recap and orientation videos, feedback, case-based discussions, and the teacher’s overall presence in the course were strategies that helped students to stay cognitively active in an asynchronous online course. The authors emphasized several strategies that promoted cognitive engagement in asynchronous online courses. These included the instructor’s active participation in discussion posts and efforts to foster meaningful dialogue. Additionally, they highlighted the importance of designing course assignments as interactive and hands-on online projects. Other strategies include conducting interviews with guest speakers on course-related topics, providing weekly recap, and orientation videos. Offering timely feedback, facilitating case-based discussions, and maintaining a strong instructor presence throughout the course were also identified as effective approaches. Together, these methods helped students stay actively engaged in the learning process.

Devlin et al. (2018) utilized BERT model to detect students’ levels of cognitive presence in the discussion posts. The existing machine learning models automatically categorizing the degree of cognitive presence (Chi and Wylie, 2014), (Hayati et al., 2020), (V. Kovanovic et al., 2016) served as an inspiration for the authors. The findings showed that depending on the course type and design, students’ cognitive presence may vary. Higher levels of cognitive presence were observed in sessions where students were asked to discuss their assignments. According to the results of the study, higher degrees of cognitive presence are strongly correlated with participants’ real final course grades, which are in line with those of Sadaf et al. (2021).

Alwafi (2022) examined how students’ cognitive presence and interactions during asynchronous discussion can be affected by elaboration feedback based on learning analytics. While the results of the first online discussion did not show any differences between the two groups,

the results of successive discussions showed that the experimental group performed better in terms of raising the levels of cognitive presence and network density. Because they were aware of the quality of their participation and the linkages between their classmates, students who received learning analytics-based elaboration feedback felt that their motivation and engagement had grown. For Graduate Communication Capstone students, this research study investigated the effects of a structured online interactive peer review system. The levels of cognitive presence were coded as exploration, triggering, integration and resolution. The authors noted that during the organized peer review process, students actively shared outside resources and cited reliable sources to support their arguments. Using the principles of cognitive apprenticeships (such as modeling, coaching, scaffolding, articulation, reflection, and exploration), this study demonstrated how computer-based cognitive tools may foster, assist, and extend learning and collaboration.

Hind et al. (2018) proposed a method to make teachers aware of students’ critical thinking and cognitive behavior so they could take the appropriate actions to ensure effective learning. This method is designed to automatically evaluate students’ CoI coded cognitive presence in relation to their social interaction in asynchronous online discussion posts. The authors analyzed student transcripts using NLP and machine learning approaches, categorizing them into five classes of cognitive presence: triggering event, exploration, integration, and resolution, and a none-category. The authors applied some standard preprocessing steps and a doc2vec feature extraction method followed by Naive Bayes classifier to classify posts into the above five classes.

Hu et al. (2021) presented a deep learning approach to automatically classify online discussion posts into CoI categories. They utilized discussion posts from two different courses and examined the generalizability and interpretability of a convolutional neural network (CNN) model for this classification problem. The performance of the CNN model was compared with random forest classifier where language aspects of psychological processes and cohesiveness had been used as features. In addition to analyzing the importance of features, the visualizations of the CNN interpretability offer unique insights into distinguishing the stages of cognitive presence in the posts.

Liu et al. (2023a) proposed MOOC-BERT which used discussion posts from MOOC to identify cognitive presence of the students. The unlabeled textual data from MOOC on various topics was used to pre-train MOOC-BERT. Authors found that MOOC-BERT considerably beat the original BERT (Yao et al., 2019), and other baseline models, such as SVM, RF, NB, DPCNN, and FastText in terms of cross-course generalizability, which indicates that it has great potential to be used in different types of courses or educational settings. The study also showed that pretraining followed by fine-tuning is even more important in situations where the model lacks high-quality labeled data, although sufficient large-scale unlabeled data are available. Additionally, the study identified performance issues in the resolution category of CoI, which need to be addressed.

Dornauer et al. (2023) developed a German-language cognitive presence classifier that was manually coded from an online university course, additional learning traces, and linguistic analysis with the Linguistic Inquiry and Word Count (LIWC) tool. Each student’s posts were classified independently into different classes of cognitive presence. The CoI coding scheme for cognitive presence (i.e., triggering event, exploration, integration, and resolution) were developed based on Garrison et al. (2001), which they translated into German and tailored to the course context. They considered various stages of cognitive presence for a post. For learning traces, they incorporated terminology from the course glossary, tagging, and attachments to posts in addition to the LIWC. Finally, they employed K-nearest neighbor, random forest model, and multilayer perceptron as the classifier. The authors in this study contributed to making the first fully CoI coded German dataset for the cognitive presence analysis.

Darabi et al. (2011) compared four different text analysis methods (Dictionary Method (DM), Coh-Metrix Tool (CMT), Interaction Analysis

Model (IAM), and CoI) and examined their effectiveness in identifying indicators of cognitive presence in online learning. The authors conducted a comparative analysis of these four methods using a dataset of asynchronous online discussions. They examined the similarities and differences in the indicators of cognitive presence identified by each method and evaluated the effectiveness of each approach. The findings of the study revealed that the CoI and the IAM model were more effective in detecting cognitive presence compared to the DM and the CMT. The CoI, in particular, provided a comprehensive framework for understanding cognitive presence by considering not only the cognitive aspects but also the social and teaching presences within online discussions.

3.3. Methods for cognitive engagement detection using KCSA framework

Hind et al. (2017) proposed a method to predict learners' level of cognitive engagement based on their involvement in online discussion posts using KCSA framework. This method combined OWL (Web Ontology Language) and LSA (Latent Semantic Analysis) ontologies and performed a semantic classification of forum posts in a particular context selected by the course instructor. To derive qualitative information from learners' forum posts on the e-learning platform, all the collected posts are analyzed using a content analysis method and coded by the cognitive dimension based on the coding scheme proposed by the KCSA. Along with this data, the authors used some numerical statistics, such as the number of posts, views, and responses with a SVM classifier to predict learners' cognitive engagement in their online courses.

Kew and Tasir (2021) examined course forum posts and analyzed how students' cognitive engagement changed over time in an online learning platform. The authors investigated the association among students' cognitive engagement, gender, and the quantity of forum posts using inferential statistics. The authors applied KCSA coding scheme to classify students' cognitive engagement into high-level and low-level involvement. Authors found that only around half of the students provided any justification for their posts, which indicated a low level of cognitive engagement. Authors emphasized the necessity to explore more options to encourage and raise students' cognitive engagement level in their course which could improve the quality of learning as well. Authors also applied Fisher's Exact Test (Upton, 1992) and Spearman's correlation and found that the level of students' cognitive engagement was unaffected by gender and a very small correlation between the number of posts and the students' cognitive engagement.

4. Discussions

4.1. Findings on research questions

Analyzing students' cognitive engagement on e-learning platforms through course discussion posts has helped researchers and instructors address a wide range of questions related to online learning. Some intriguing research questions and their potential solutions, based on existing literature, are discussed below:

RQ 1: What is the level of cognitive engagement among students in online learning discussion forums? The level of cognitive engagement among students in online learning discussion forums varies depending on several factors, including the design of the course, the quality of the discussion prompts, and the level of student participation. Research suggests that students may engage in different types of cognitive activities in online discussion forums, including elaborating on ideas, questioning assumptions, and analyzing information. However, a study by Kew and Tasir (2021) has found that students may also engage in superficial or low-level discussions, such as summarizing information or expressing agreement or disagreement with others. To improve the level of cognitive engagement in online learning discussion forums, instructors can use strategies such as providing clear and open-ended prompts, modeling high-level discussion, and encouraging active participation from all students (Kew and Tasir, 2021).

RQ2: How does the level of cognitive engagement differ between students with different demographic characteristics (e.g., age, gender, academic background)? The level of cognitive engagement in online learning may differ based on various demographic characteristics of students, such as age, gender, and academic background. Richardson and Newby (2006) found that age is a significant predictor of cognitive engagement in online learning, with older students exhibiting higher levels of engagement compared to younger students. However, another study (Park et al., 2014) found no significant differences in cognitive engagement based on age. Gender has also been studied as a potential predictor of cognitive engagement in online learning, where Hartono et al. (2019) found that female students exhibited higher level of engagement compared to male students, while others have found no significant differences based on gender. Moreover, academic background has also been investigated as a potential predictor of cognitive engagement in online learning. Kew and Tasir (2021) found that students with a higher level of academic achievement or experience in the subject matter exhibit higher levels of engagement compared to students with lower levels of academic achievement or experience.

RQ3: How does the level of cognitive engagement change over time throughout the course? The level of cognitive engagement in online learning can change over time throughout the course, influenced by various factors such as the course design, the instructional strategies used, and the level of student motivation. Singh et al. (2022) investigated changes in cognitive engagement over time in online courses and found varying results. Singh et al. (2022) has found that cognitive engagement tends to be higher at the beginning of the course when students are more motivated and excited about the new learning experience. However, as the course progresses, cognitive engagement tends to decrease due to factors such as boredom, frustration, or lack of engagement with the course material or activities. Richardson and Newby (2006) found that cognitive engagement may vary depending on the specific course activities or modules being studied. For example, students may exhibit higher levels of engagement during more interactive or challenging activities, while engagement may decrease during more passive or repetitive activities. Gray and DiLoreto (2016) also stated that the level of cognitive engagement in online learning is dynamic and may change over time throughout the course. To promote sustained engagement, course designers and instructors may need to incorporate a variety of instructional strategies and activities that challenge and motivate students throughout the course, while also providing opportunities for reflection and feedback. Additionally, regular communication and support from the instructor can also help to maintain students' cognitive engagement over time.

RQ4: What types of cognitive engagement are most prevalent among students in online learning discussion forums? Online learning discussion forums can provide students with opportunities to engage in various types of cognitive activities, such as reflection, critical thinking, and problem-solving (Hind et al., 2017). Several studies have investigated the types of cognitive engagement that are most prevalent among students in online learning discussion forums. One type of cognitive engagement that is often observed in online learning discussion forums is reflective thinking. Students may engage in reflective thinking by sharing personal experiences, making connections between course concepts and real-world situations, and discussing their learning processes and progress. Another type of cognitive engagement commonly observed in online learning discussion forums is critical thinking. Students may engage in critical thinking by analyzing and evaluating course concepts, questioning assumptions, and providing evidence to support their arguments or perspectives. Problem-solving is also a common type of cognitive engagement observed in online learning discussion forums. Students may engage in problem-solving by applying course concepts to real-world problems, collaborating with peers to find solutions, and evaluating the effectiveness of different solutions.

RQ5: How do different types of cognitive engagement relate to student learning outcomes (e.g., grades, knowledge retention, satisfaction)?

Several studies have shown a positive relationship between cognitive engagement and student learning outcomes in online learning environments (Richardson, 2001), (Li et al., 2023), (Gray and DiLoreto, 2016), (Cho and Heron, 2015). For example, Li et al. (2023) found that higher levels of cognitive engagement, particularly in the areas of critical thinking and higher-order thinking, were positively correlated with higher grades and better performance on exams. Richardson (2001) found that higher levels of cognitive engagement, specifically in the areas of critical thinking and reflective thinking, were positively related to student satisfaction with the course and their perceived learning outcomes. Similarly, Cho and Heron (2015) found that higher levels of cognitive engagement, particularly in the areas of critical thinking and self-regulated learning, were positively correlated with knowledge retention. These studies suggest that promoting higher levels of cognitive engagement among students in online learning environments can lead to improved learning outcomes, including higher grades, better exam performance, increased satisfaction with the course, and better retention of knowledge.

RQ6: How does the level of cognitive engagement differ across different types of online learning activities (e.g., discussion forums, quizzes, group projects)? The level of cognitive engagement can differ across different types of online learning activities, as each type of activity may require different cognitive processes or may be conducive to certain types of engagement. Discussion forums are often used to promote reflective thinking, critical thinking, and social learning. Students may engage in a variety of cognitive processes during forum discussions, including analyzing, evaluating, and synthesizing information, as well as sharing personal experiences and perspectives. Therefore, discussion forums may be particularly conducive to cognitive engagement that promotes deep learning and knowledge construction. Quizzes are often used to promote retrieval practice, which can enhance memory retention and knowledge acquisition. However, the cognitive engagement required during quizzes may be more focused on recall and recognition rather than higher-order thinking or knowledge construction. Another important aspect is group projects. Group projects can promote collaboration, communication, and problem-solving skills (Kim, 2013). The cognitive engagement required during group projects may be more focused on applying knowledge to real-world problems, generating new ideas, and synthesizing information from multiple sources. Overall, the level and type of cognitive engagement required during different types of online learning activities can vary widely, and each type of activity may have unique strengths and limitations for promoting cognitive engagement and learning outcomes. Understanding the specific types of cognitive engagement required for each activity can help instructors design more effective online learning experiences and promote deeper levels of cognitive engagement among students.

RQ7: How do different instructional designs and pedagogical strategies impact cognitive engagement in online learning? Different instructional designs and pedagogical strategies can significantly impact cognitive engagement in online learning (V. Kovanovic et al., 2016). For example, instructional designs that incorporate active learning strategies such as problem-based learning, case-based learning, and collaborative learning have been found to enhance cognitive engagement. These strategies encourage students to take an active role in their learning, work collaboratively with peers, and apply their knowledge to real-world problems. Similarly, pedagogical strategies such as providing timely feedback, setting clear learning goals, and using multimedia resources can also promote cognitive engagement. Providing timely feedback can help students understand their progress and identify areas where they need to improve. Setting clear learning goals can help students stay focused and motivated while using multimedia resources can provide a variety of engaging learning experiences. On the other hand, instructional designs that rely heavily on lecture-based formats and passive learning strategies have been found to limit cognitive engagement. These formats often prioritize the delivery of information

overactive engagement and fail to provide opportunities for students to apply their knowledge.

RQ8: How can instructors use cognitive engagement data to optimize their teaching strategies and improve student learning outcomes? Instructors can use cognitive engagement data to optimize their teaching strategies and improve student learning outcomes in several ways (Ozogul et al., 2022). By analyzing cognitive engagement data, instructors can identify areas where students may be struggling or disengaged and need improvement. This information can help instructors adjust their teaching strategies to better meet the needs of their students. Cognitive engagement data can also help instructors provide targeted feedback to students. For example, if students are struggling with a particular concept or activity, instructors can provide personalized feedback to help students improve. In addition, based on cognitive engagement data, instructors can adjust their instructional strategies to better engage students. For example, if students are disengaged during lectures, instructors can incorporate more interactive activities to increase engagement. Cognitive engagement data can also be used to personalize learning experiences for students. Instructors can use data to identify individual student strengths and weaknesses and provide tailored learning experiences that better meet their needs. By monitoring cognitive engagement data over time, instructors can track student progress and adjust as needed to ensure students are meeting learning objectives. Overall, cognitive engagement data can provide valuable insights for instructors to optimize their teaching strategies and improve student learning outcomes.

4.2. Scalability challenge in cognitive engagement frameworks

The ICAP framework demonstrates scalability, particularly in data-driven environments such as online discussion forums, MOOCs, and large-scale asynchronous platforms. Its strength lies in its behavioral taxonomy, which maps well onto automated classification techniques using machine learning and NLP. For example, Hayati et al. (2020) applied text mining and SVM classifiers to effectively detect ICAP engagement levels in an online software engineering course. Similarly, Liu et al. (2022) utilized the BERT-CNN model to classify cognitive and emotional engagement in MOOC forums, showing robust results at scale. The framework's categorical structure (i.e., Passive, Active, Constructive, Interactive) allows researchers to annotate and automate classification pipelines with a relatively high degree of reliability. However, ICAP has certain limitations. The framework predominantly captures observable behavioral engagement, potentially overlooking deeper cognitive or metacognitive dimensions such as reflection or emotion-driven learning. Additionally, high-fidelity classification still relies on expert-labeled datasets for training, which may not generalize well across varying disciplines or cultures. Therefore, while ICAP is scalable across large textual data sources, such as forum posts or chat logs, it may be less effective in multimodal or nuanced instructional contexts unless further adapted.

The CoI framework is inherently scalable across diverse educational contexts, including higher education, K–12, and blended or fully online courses. Its tripartite structure—comprising cognitive presence, social presence, and teaching presence—has been validated in numerous studies as a comprehensive lens for evaluating learning engagement. Lee et al. (2022) applied a BERT-based model to automatically classify forum posts into CoI categories, demonstrating the framework's applicability to large corpora. Similarly, Liu et al. (2023a) developed MOOC-BERT, a domain-specific language model trained on unlabeled MOOC discussions and achieved high cross-course generalizability for detecting cognitive presence. Despite its scalability, CoI presents some challenges when applied to a complex or high-volume environment. The framework's nuanced dimensions require substantial annotation effort and domain knowledge, which can hinder rapid deployment without significant computational or human resources. Furthermore, reliance on self-reported data in some implementations can limit its

accuracy, and its application may be less suited to non-traditional learning platforms where structured interactions (e.g., instructor facilitation) are minimal or absent. Nonetheless, with proper infrastructure and automation, CoI remains a powerful framework for large-scale analysis of cognitive engagement.

The KCSA framework offers a detailed, cognitively focused structure for analyzing student discussions. It segments learning into key phases: exploration, conceptualization, application, and reflection. For example, Hind et al. (2017) and Kew and Tasir (2021) used KCSA to classify cognitive engagement levels using both statistical and semantic methods (e.g., OWL, LSA, and SVM) on forum post data. These implementations show the framework's potential for scale, especially in well-structured LMS environments or instructor-moderated forums. However, KCSA's granularity and complexity can limit its scalability across broader or less structured datasets. The coding scheme involves multiple subcategories (e.g., CHV1, C1I, C1E, REJECT+), which require intensive manual annotation or sophisticated automation. Unlike ICAP or CoI, KCSA has seen less adoption in large-scale MOOC contexts, and its application beyond forum-based environments is relatively limited. Therefore, while KCSA is valuable for detailed cognitive analysis, particularly in small to medium-sized courses, it may not yet be optimized for real-time or large-scale adaptive learning environments.

4.3. Domain specificity and context sensitivity

Domain specificity is an essential factor in assessing the effectiveness of cognitive engagement models in various learning environments. The ICAP framework provides a broad range of usefulness, beneficial for various fields such as STEM, social sciences, business, and humanities. Its hierarchical framework for engagement enables researchers to categorize involvement across a wide array of online learning environments. However, ICAP is limited in areas where real-time collaboration is needed, as it mainly concentrates on individual student actions rather than on the shared construction of knowledge.

The CoI framework is most appropriate for educational settings that involve discussions, reflective dialogue, argumentation, and collaborative knowledge building. It is most effective in the humanities and social sciences, where instructor guidance (teaching presence) and student interaction (social presence) play crucial roles in participation. Nevertheless, CoI is not as effective in STEM disciplines, where engagement in learning typically occurs through experiments, problem-solving tasks, and coding exercises instead of text-focused discussions. Consequently, CoI does not consistently indicate participation in highly interactive or simulation-driven learning settings (Akyol and Garrison, 2011).

The KCSA framework is very specific to certain domains, with its most efficient uses found in areas like law, business, and political science that rely on argumentation. Its emphasis on cognitive processing, problem-solving, and decision-making makes it particularly well-suited for group project work and debate courses. It is less suitable in technical fields like computer science or engineering, where participation is typically shown through practical work instead of written communication. Additionally, the model does not align with self-paced learning models, resulting in its unsuitability for adaptive learning platforms (Vos et al., 2011).

4.4. Adaptability to evolving AI-driven online learning environments

The adaptability of cognitive engagement frameworks to modern AI-driven learning environments is crucial for maintaining their ongoing relevance. The ICAP framework has been widely used in machine learning-based engagement detection, with models such as SVM, Random Forest, and neural networks successfully classifying ICAP-based engagement levels. However, ICAP does not account for emotional and social engagement, which are significantly relevant in modern AI-driven learning. Moreover, its applicability is limited to text-based interactions, making it difficult to integrate with video-based learning,

multimodal interactions, and real-time adaptive learning systems (Chi and Boucher, 2023), (Thurn et al., 2023).

The CoI framework has shown some adaptability to modern NLP-based analysis, with studies employing BERT-CNN models, sentiment analysis, and deep learning classifiers to evaluate cognitive presence. However, its strong reliance on discussion-based interactions limits its applicability to self-paced learning environments, where learners primarily engage with AI tutors rather than human peers. Additionally, CoI faces challenges with multimodal engagement detection, as it was not designed to incorporate speech, visual cues, or behavioral data (Garrison et al., 1999), (Akyol and Garrison, 2011).

The KCSA framework encounters the most significant challenges in adaptability. Unlike ICAP and CoI, which have been integrated into NLP-driven engagement detection models, KCSA's dependence on structured argumentation and decision-making processes makes it difficult to automate. Furthermore, it does not align well with real-time engagement tracking systems, making it unsuitable for modern AI-driven adaptive learning environments (Vos et al., 2011).

4.5. Cognitive engagement detection models: Strengths and weaknesses

The identification of cognitive engagement in online education has greatly advanced through the integration of NLP and Deep Learning (DL) methods. Conventional techniques, like rule-based language analysis and statistical NLP systems, have gradually been replaced by transformer-based frameworks and neural embedding models that provide enhanced contextual understanding and scalability. Early engagement detection methods mainly depended on lexicon-based strategies like LIWC and Coh-Metrix, which examine sentiment, lexical variety, and cohesion in discussion contributions. Although these techniques provide an interpretable way to evaluate engagement, they face challenges in scalability and adaptability in various learning settings because they depend on predefined dictionaries and linguistic rules. In the same way, classifiers based on machine learning, such as SVM, RF, and Logistic Regression (LR), have been utilized for engagement classification by utilizing manually created linguistic and contextual features. While these models provide improved accuracy over lexicon-based methods, they require significant feature engineering and do not fully grasp more profound cognitive engagement patterns beyond surface text analysis.

The introduction of deep learning models, especially transformer-based architectures like BERT, GPT, and XLNet, has greatly enhanced engagement detection by identifying hidden cognitive engagement patterns and contextual dependencies in conversations. These models, pre-trained on large corpora, exhibit excellent generalizability across various learning environments, making them suitable for ICAP, CoI, and KCSA frameworks. Nevertheless, despite their enhanced accuracy, deep learning models present computational difficulties, requiring large datasets and high processing power, which constrains their feasibility for real-time engagement monitoring in online learning settings. To tackle these challenges, hybrid AI models have emerged, blending conventional NLP heuristics with neural embeddings. For example, the BERT-CNN and BERT-LIWC models integrate deep contextual embeddings with structured knowledge frameworks to improve both interpretability and computational efficiency. These hybrid methods surpass individual models by providing enhanced generalization and decreasing overfitting to discussion-specific language.

5. Conclusion

The ICAP, CoI, and KCSA frameworks have been found to be important in detecting cognitive engagement of students in online learning. These frameworks can serve cognitive engagement detection methods in several ways. The analysis and categorization of student behavior during online learning using these frameworks offers better organization, which makes it simpler for instructors to comprehend and evaluate

the data. By considering several aspects of learning, such as social presence, cognitive presence, and specific categories of cognitive processes, these frameworks provide a holistic view of student participation. This gives instructors insights into how the students interact with the information and activities in the courses. These frameworks allow instructors to gather information on student behavior and utilize it to support data-driven decisions on how to improve the learning outcomes for their students. Instructors can optimize the learning outcomes for their students by analyzing how well their classes are using these frameworks. This covers adjustments to learning materials, teaching strategies, and course design. These frameworks can assist instructors in making sure that their teaching methods complement their intended learning outcomes. Instructors may make sure they are fostering a learning environment that is optimized for student success by focusing on forms of engagement or aspects of learning.

These frameworks offer a structured, thorough, and data-driven approach to understanding student behavior, making them crucial in the detection of cognitive engagements of students during online learning. This can result in more effective teaching strategies and better learning outcomes for students. While the frameworks are useful for detecting cognitive engagement of students during online learning, there are some limitations and factors that may affect their accuracy and applicability. There are some instances when these frameworks may fail in detecting cognitive engagement. These frameworks focus on the examination of text-based data from written assignments, chat logs, and discussion forum entries. They might not fully capture the interaction's environment, which can affect students' cognitive engagement through nonverbal clues, social dynamics, and cultural considerations. The coding categories of this framework may be arbitrary and subject to different interpretations. As a result, the classification of engagement types and the coding procedure may become inconsistent. These frameworks may not account for all forms of cognitive involvement, including metacognitive, creative, and emotional engagement. As a result, they might give an inaccurate image of how students are engaging cognitively in online learning environments. These frameworks heavily rely on NLP and machine learning techniques to analyze large amounts of text data. However, the accuracy of NLP and machine learning techniques may vary depending on the quality and complexity of the text data, which can affect the accuracy of cognitive engagement detection. Individual variations in cognitive engagement styles and preferences might not be considered by these frameworks. For instance, some students could like in-depth reflection, while others might favor learning activities that require greater participation and activity. Overall, while these frameworks can be valuable tools for detecting the cognitive engagement of students during online learning, it is important to consider their limitations and potential biases to ensure that they provide an accurate and comprehensive understanding of students' learning experiences.

It is important to note that each framework has its strengths and weaknesses, and the choice of the framework should be based on the research question and the educational context. From different discussions, we extracted the advantages and disadvantages of each framework in detecting cognitive engagement in online learning. The ICAP framework is accessible to a broad spectrum of academics and educators due to its simplicity of use and comprehension. It has been verified in a variety of educational environments and is concentrated on the direct observation of learning behaviors, making it valuable for identifying areas where interventions may be required. However, it only looks at cognitive engagement and ignores other elements that might affect student learning, like motivation or emotional involvement. ICAP also needs skilled observers to implement it, which would prevent it from being widely used. It could not be sensitive enough to pick up on slower, more gradual changes in cognitive activation.

Contrarily, the CoI framework views student participation holistically, considering the interactions between teaching presence, social presence, and cognitive presence. It has undergone significant research

and has been approved in numerous educational settings. It can be used practically to direct the creation and administration of online courses. However, CoI can be difficult to implement and complex, necessitating a thorough comprehension of the framework and its elements. This paradigm mainly relies on data that was self-reported, which does not always correctly reflect student participation. Furthermore, it might not be a good tool for evaluating participation in other online learning contexts, like MOOCs.

A significant part of student learning is knowledge building, which is the focus of the KCSA framework. It has received validation in numerous diverse educational environments. It is relatively simple to use and put into practice. The methodology exclusively evaluates knowledge construction; it does not include other facets of student interaction. The researcher or instructor must exercise some degree of subjectivity to evaluate the data. Additionally, it might not be a good way to gauge participation in other online learning environments, like MOOCs. The framework has the advantage of offering a methodical technique to assess the cognitive processes that students use during online discussions. Educators and researchers can create interventions to support and enhance students' cognitive engagement and foster deeper learning by determining the stage of cognitive processing that students are in. However, one potential drawback of the framework is that using the coding scheme to analyze online discussions is time-consuming. Furthermore, the framework might fall short of capturing the dynamic and complex nature of online discussions, which can involve several stages of cognitive processing at once.

The ICAP, CoI, and KCSA frameworks have been used in this review paper to shed light on the challenging task of identifying cognitive engagement among online learners. Our investigation has highlighted cognitive engagement's multidimensional nature and stressed how crucial it is to create successful online learning environments. The fusion of deep learning and machine learning methods has ushered in a new era of automatic engagement detection, potential targeted interventions, and improved instructional design. Although the frameworks have given a thorough foundation, difficulties still exist. Continued research is required due to the demand for uniform metrics, ethical issues with data utilization, and thorough cross-framework analysis. The combination of these frameworks offers a chance for a more comprehensive method of detecting cognitive involvement, enabling teachers to design precise interventions that consider various learning preferences.

In the future, the dynamic nature of online learning will necessitate models that are adaptable and take changing engagement patterns into consideration. Accurate detection may be improved by including multimodal data, such as visual and behavioral clues. Additionally, real-time feedback methods can establish a feedback loop that dynamically modifies instructional strategies while also informing engagement. Thorough knowledge of cognitive engagement might also result from research into hybrid models, which combine the advantages of many frameworks. A cohesive paradigm that incorporates the cognitive, social, and emotional facets of involvement may result from this. It is also important to continue paying attention to ethical issues, particularly those involving learner autonomy and data privacy. It is crucial to strike a balance between raising engagement and protecting people's privacy. This review essentially emphasizes the need for cognitive engagement in enabling successful online learning. We foresee a future where automated detection seamlessly influences instructional tactics, providing an educational environment that is engaging, responsive, and adaptive by fusing theoretical frameworks with cutting-edge technologies.

CRedit authorship contribution statement

Nazmus Sakeef: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **M. Ali Akber Dewan:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Fuhua Lin:** Writing – review & editing, Validation, Investigation, Conceptualization. **Dharamjit Parmar:** Writing – review & editing, Methodology, Investigation, Conceptualization.

Declaration of competing interest

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